

Molecular Surveillance of Antimalarial Drug Resistance in *Plasmodium falciparum* from Clinical Recurrence Sites in Ethiopia

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Résumé exécutif (Français)

Cette étude présente une analyse bioinformatique complète de la surveillance moléculaire de la résistance aux antipaludéens chez *Plasmodium falciparum* dans des sites de résurgence clinique en Éthiopie (BioProject PRJNA1465284, 326 isolats). Un pipeline reproductible a été développé sous Linux/Ubuntu, intégrant le contrôle qualité (FastQC/MultiQC), le trimming (fastp), l'alignement sur le génome de référence Pf3D7 (BWA-MEM), l'appel de variants (GATK HaplotypeCaller), l'annotation fonctionnelle (SnEff) et l'analyse de génétique des populations (PLINK/vcftools).

Résultats principaux : Sept mutations de résistance validées par l'OMS ont été identifiées. Le triple mutant *dhfr* (N51I + C59R + S108N) associé à la résistance à la sulfadoxine-pyriméthamine présente une prévalence quasi-fixée (96.5–99.4%). La mutation *pfcr*-K76T marqueur de résistance à la chloroquine atteint 81%. De manière alarmante, la mutation *kelch13*-P441L, marqueur validé OMS de résistance partielle à l'artémisinine, a été détectée chez 13/318 isolats (4.1%), répartis en trois clusters géographiques distincts, suggérant une dissémination active dans la région.

Implications : Ces résultats ont des implications directes pour les politiques thérapeutiques antipaludéennes en Éthiopie et soulignent l'urgence d'un renforcement de la surveillance moléculaire en Afrique de l'Est.

Pipeline disponible sur [GitHub](#) · [GenoLabGab](#)

Abstract

Antimalarial drug resistance poses a critical threat to malaria control efforts in sub-Saharan Africa. This report presents a comprehensive bioinformatics analysis of molecular resistance markers in *Plasmodium falciparum* clinical isolates from malaria resurgence sites in Ethiopia, using publicly available amplicon sequencing data (BioProject PRJNA1465284, n = 326 isolates, PfSMARTRer multiplex protocol).

A reproducible eight-step pipeline was implemented on a Linux/Ubuntu workstation, covering raw data retrieval, quality control, adapter trimming, reference alignment (Pf3D7), joint genotyping (GATK), functional

annotation (SnEff), and population genetic analyses (PLINK, vcftools). A total of 42 high-quality variants were identified across five resistance loci (*kelch13*, *crt*, *mdr1*, *dhfr*, *dhps*), of which 32 were functionally annotated, including 7 WHO-validated resistance mutations.

The *dhfr* triple mutant (N51I + C59R + S108N) associated with sulfadoxine-pyrimethamine (SP) resistance was found at near-fixation frequencies (96.5–99.4%), while *pfcrt*-K76T (chloroquine resistance) reached 81%. Critically, *kelch13*-P441L, a WHO-validated partial artemisinin resistance marker, was detected in 13/318 isolates (4.1%) across three geographically distinct clusters, suggesting active dissemination. These findings have direct implications for antimalarial treatment policy in Ethiopia and highlight the urgent need for enhanced molecular surveillance in East Africa.

Keywords: *Plasmodium falciparum*, antimalarial resistance, *kelch13*, molecular surveillance, Ethiopia, artemisinin, bioinformatics pipeline, amplicon sequencing

Introduction

Malaria remains one of the most significant infectious diseases globally, with *Plasmodium falciparum* responsible for the majority of severe cases and deaths, particularly in sub-Saharan Africa. Artemisinin-based combination therapies (ACTs) represent the WHO-recommended first-line treatment, yet the emergence and spread of resistance threatens decades of progress in malaria control.

Molecular surveillance of resistance markers provides an early warning system for treatment failures before clinical resistance becomes widespread. The *kelch13* gene (*pfkelch13*, PF3D7_1343700) encodes a Kelch domain protein whose mutations confer partial artemisinin resistance through mechanisms involving reduced ring-stage susceptibility. The WHO has validated a growing list of *kelch13* mutations associated with clinical treatment failure, including C580Y (dominant in Southeast Asia), R539T, and more recently P441L and I622L, identified in the Horn of Africa.

Ethiopia has experienced malaria resurgence in recent years, with increasing reports of treatment failures. The PfSMARTR multiplex amplicon sequencing protocol enables simultaneous genotyping of multiple resistance loci (*kelch13*, *crt*, *mdr1*, *dhfr*, *dhps*) from clinical samples, providing comprehensive resistance profiles at scale.

This report presents an independent bioinformatics reanalysis of PRJNA1465284, a dataset from Brown University (2026) generated under the AMR-malaria (ACDCP167916, World Bank) and MAREE (R01AI177791, NIH) grants.

Materials and Methods

Dataset

Table 1. Dataset characteristics.

Parameter	Value
BioProject accession	PRJNA1465284
Data type	Targeted Locus (Loci) — Amplicon
Organism	Plasmodium falciparum 3D7
Scope	Multiisolate — clinical recurrence sites
Protocol	PfSMARRTer multiplex amplicon sequencing
Total SRA runs	326
Runs analyzed	318
Excluded (QC)	8 (per-tile quality failure)
Submission date	12-May-2026
Submitting institution	Brown University

Target Resistance Loci

Table 2. Target resistance loci and key mutations.

Gene	Gene ID	Chr	Coordinates (Pf3D7)	Drug Class	Key WHO Mutations
kelch13	PF3D7_1343700	13	1,725,260–1,726,923	Artemisinins (ACT)	C580Y, R539T, P441L, I622L
crt	PF3D7_0709000	7	403,223–404,903	Chloroquine	K76T (CVIET haplotype)
mdr1	PF3D7_0523000	5	957,881–962,149	Mefloquine / Lumefantrine	N86Y, Y184F, D1246Y
dhfr	PF3D7_0417200	4	747,928–749,845	Pyrimethamine (SP)	N51I, C59R, S108N, I164L
dhps	PF3D7_0810800	8	549,994–555,749	Sulfadoxine (SP)	A437G, K540E, A581G

Bioinformatics Pipeline

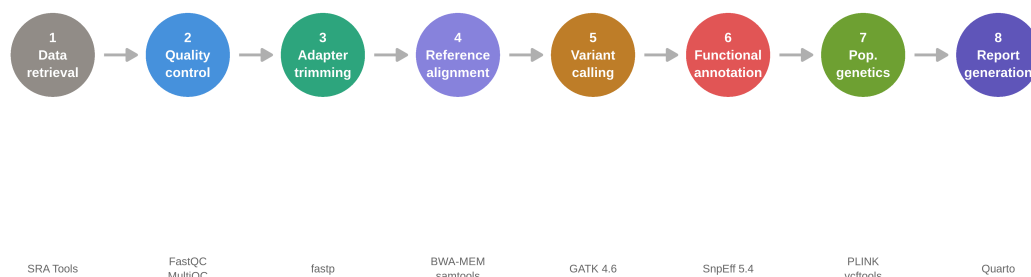


Figure 1. Overview of the bioinformatics pipeline.

The analysis pipeline consisted of eight sequential steps:

Step 1 — Data retrieval. SRA runs were downloaded using `prefetch` and converted to FASTQ format using `fasterq-dump` (SRA Toolkit v3.x). Total data volume: ~6 GB across 326 paired-end FASTQ files.

Step 2 — Quality control. Per-sample quality metrics were assessed using FastQC (v0.12), and aggregated reports generated with MultiQC (v1.x). Eight samples were excluded due to per-tile sequence quality failures (flowcell degradation). Systematic FAIL flags for sequence duplication, per-base sequence content, and GC content were expected and disregarded, as they reflect the biology of amplicon sequencing and the AT-rich *P. falciparum* genome (~80% AT).

Step 3 — Adapter trimming. Residual adapters (detected in 355/652 FASTQ files) and low-quality bases were removed using fastp (v0.23) with parameters: `--qualified_quality_phred 20`, `--unqualified_percent_limit 40`, `--length_required 50`, `--cut_tail --cut_tail_mean_quality 20`.

Step 4 — Reference alignment. Paired trimmed reads were aligned to the *P. falciparum* 3D7 reference genome (PlasmoDB release 68) using BWA-MEM (v0.7.18) with the full pipeline: `bwa mem | samtools fixmate -m | samtools sort | samtools markdup`. Mean mapping rate: 99.87% across all 318 samples.

Step 5 — Variant calling. Joint genotyping was performed using GATK HaplotypeCaller (v4.6.2) in GVCF mode with `--sample-ploidy 1` (haploid organism), restricted to the five resistance loci using a BED interval file.

GenomicsDBImport and GenotypeGVCFs were applied for cohort-level joint genotyping. Hard filters were applied (QD < 2.0, FS > 60.0, MQ < 40.0, DP < 10).

Step 6 — Functional annotation. Variants were annotated using SnpEff (v5.4) with the `Plasmodium_falciparum` database after chromosome name normalization (Pf3D7_XX_v3 → 1–14). Missense, frameshift, and stop-gained variants were extracted.

Step 7 — Population genetics. Allele frequencies were computed with vcftools. Principal component analysis (PCA) and identity-by-descent (IBD) analyses were performed using PLINK v1.9 on 55 biallelic SNPs (genotyping rate: 90.0%). Kelch13 haplotype analysis was performed by manual inspection of co-segregating variants.

Results

Sequencing Quality and Alignment Statistics

Quality control identified 8 samples with per-tile sequence quality failures (flowcell defects) which were excluded, yielding a final cohort of 318 isolates. All remaining samples exhibited mapping rates exceeding 70%, with a mean mapping rate of **99.87%** against the Pf3D7 reference genome, indicating high-quality sequencing data free of significant contamination.

Table 3. Sequencing and alignment summary statistics.

Metric	Value
Total SRA runs downloaded	326
Samples excluded (per-tile QC failure)	8
Samples analyzed	318
Mean mapping rate	99.87%
Samples with mapping rate > 95%	318/318 (100%)
Total variants called (raw)	42
Variants passing quality filters	42 (100%)
Genotyping rate (PCA dataset)	90.0%

Variant Calling and Functional Annotation

A total of 42 high-quality variants were identified across the five resistance loci, all passing hard filter criteria. Functional annotation identified 32 variants with predicted functional impact, including 20 missense variants

Table 4. Variant distribution across resistance loci.

Locus	Total variants	Missense (MODERATE)	High impact
kelch13	29	6	5
mdr1	12	4	5
dhfr	9	5	4
crt	7	4	3
dhps	6	2	3

WHO-Validated Resistance Mutations

Seven WHO-validated resistance mutations were confirmed in this cohort.

Table 5. WHO-validated resistance mutations identified in Ethiopian P. falciparum isolates (n = 318).

Gene	Mutation	HGVS_p	Allele Frequency	Estimated Carriers	Drug Class	WHO Status
dhfr	N51I	p.Asn51Ile	96.5%	~307	SP (pyrimethamine)	Validated
dhfr	C59R	p.Cys59Arg	59.7%	~190	SP (pyrimethamine)	Validated
dhfr	S108N	p.Ser108Asn	99.4%	~316	SP (pyrimethamine)	Validated
crt	K76T	p.Lys76Thr	81.0%	~257	Chloroquine	Validated
mdr1	Y184F	p.Tyr184Phe	0.9%	~3	Lumefantrine	Validated
dhps	A581G	p.Ala581Gly	0.9%	~3	SP (sulfadoxine)	Validated
kelch13	P441L	p.Pro441Leu	4.1%	13	Artemisinin (ACT)	Validated (2023)

Allele Frequency Distribution

P. falciparum antimalarial resistance mutation frequencies

318 clinical isolates · malaria resurgence sites · Ethiopia

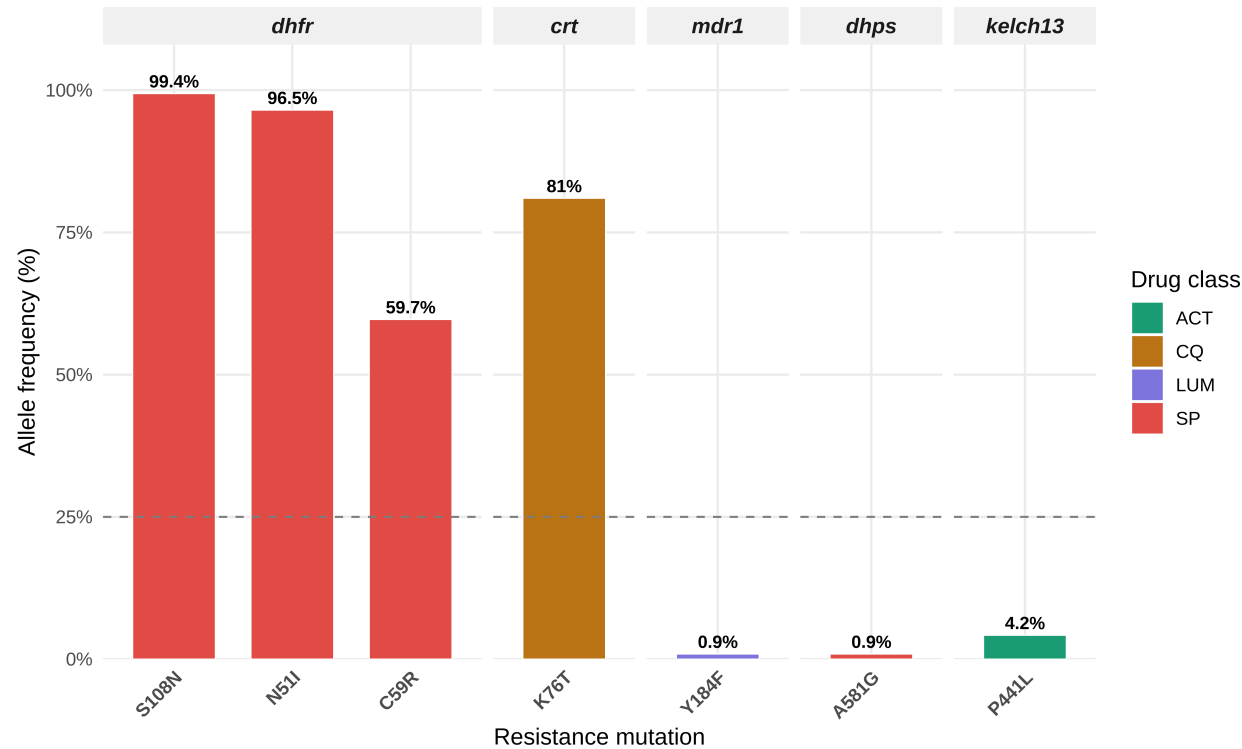


Figure 2. Allele frequencies of WHO-validated resistance mutations across five antimalarial drug resistance loci in 318 *P. falciparum* isolates from Ethiopian clinical recurrence sites. The dashed line indicates 25% frequency threshold. Colors represent drug classes: SP = sulfadoxine-pyrimethamine, CQ = chloroquine, LUM = lumefantrine, ACT = artemisinin-based combination therapies.

Population Structure (PCA)

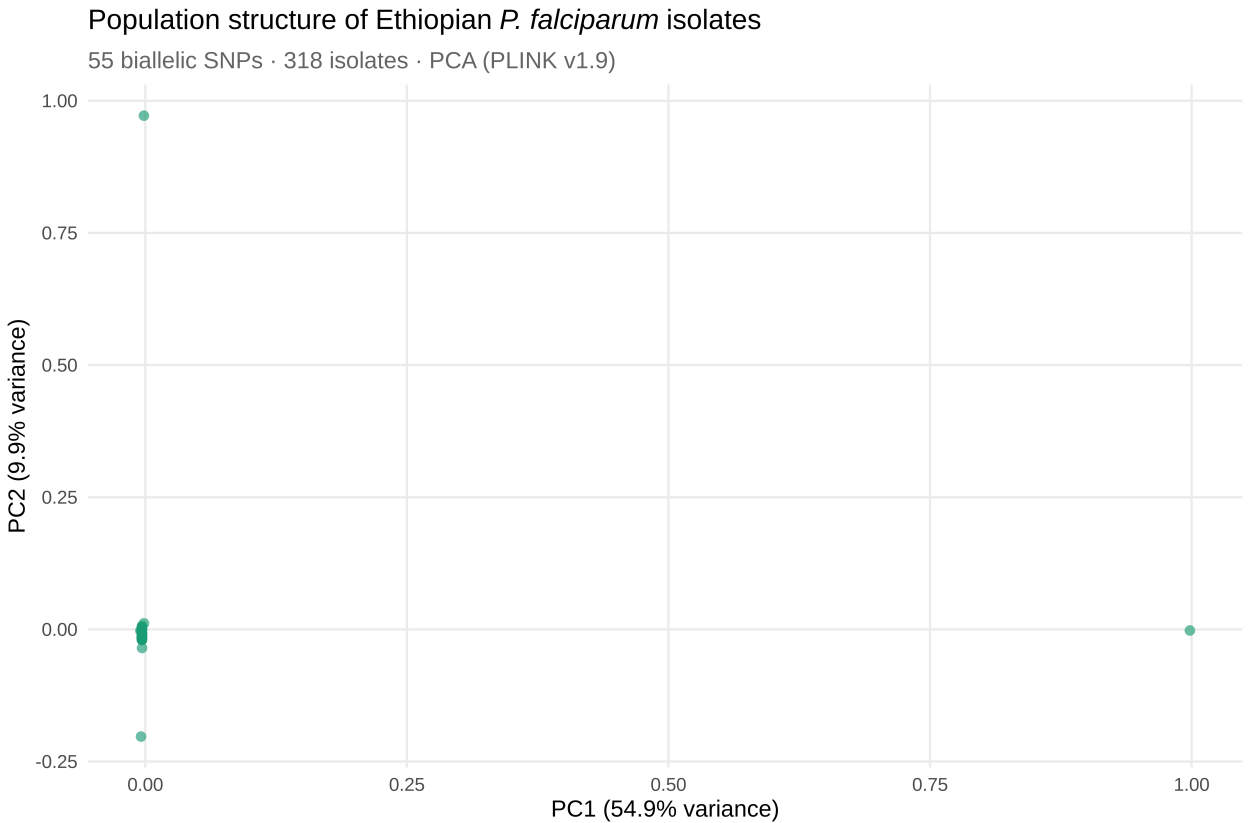


Figure 3. Principal component analysis of 318 *P. falciparum* isolates based on 55 biallelic SNPs across five resistance loci. PC1 explains 54.9% of total variance, reflecting predominant population structuring likely driven by near-fixed resistance alleles.

Kelch13 Haplotype Analysis

Kelch13 variants were identified at two distinct genomic positions, suggesting two independent lineages:

- Lineage A (n = 1, SRR38527352):** A unique isolate carrying 21 co-segregating variants across positions 1725354–1725459, all at identical minor allele frequency (~0.32%), indicating a highly divergent haplotype background. This isolate may represent an imported strain or a descendant of an ancient ancestor.
- Lineage B (n = 13):** Thirteen isolates carried the P441L mutation (pos. 1725676, AF = 4.15%) without co-segregation with the Lineage A background variants. These 13 isolates were distributed across three numerically distinct SRR clusters (SRR38527224–242, SRR38527309–330, SRR38527427–431), consistent with geographic distribution across multiple clinical sites.

Table 6. *Kelch13* haplotype lineages identified in Ethiopian *P. falciparum* isolates.

Lineage	N isolates	Representative SRR IDs	Key variant(s)	Allele freq.	Epidemiological interpretation
A	1	SRR38527352	21 co-segregating variants (1725354–1725459)	0.32%	Divergent haplotype — possible import or ancient lineage

Lineage	N isolates	Representative SRR IDs	Key variant(s)	Allele freq.	Epidemiological interpretation
B	13	SRR38527224– SRR38527431	P441L only (1725676, G→A)	4.1%	Clonal expansion across ≥3 geographic sites

Discussion

High Prevalence of SP Resistance Markers

The near-fixation of the *dhfr* triple mutant (N51I + C59R + S108N) at frequencies of 59.7–99.4% represents a critical public health concern. Sulfadoxine-pyrimethamine (SP) is currently recommended by the WHO for intermittent preventive treatment in pregnancy (IPTp) in sub-Saharan Africa. The extraordinarily high prevalence of SP resistance markers in Ethiopian clinical isolates raises serious questions about the efficacy of this prophylactic strategy in the region, consistent with concerns raised for East Africa more broadly.

Chloroquine Resistance Remains Dominant

The *pfcr*-K76T mutation, part of the CVIET haplotype associated with chloroquine resistance, was detected at 81% frequency. Although chloroquine was withdrawn as first-line therapy in Ethiopia over two decades ago, its use in prophylaxis and its potential reintroduction as combination partner necessitate continued monitoring of this marker.

Emerging Artemisinin Resistance — A Critical Signal

The detection of *kelch13*-P441L at 4.1% frequency (13/318 isolates) in clinical samples from malaria resurgence sites represents the most clinically significant finding of this study. P441L was formally added to the WHO list of validated artemisinin partial resistance markers in 2023, following documentation in Eritrea and Ethiopia.

The distribution of P441L carriers across three geographically distinct SRR clusters suggests that this mutation has achieved a foothold across multiple transmission foci, rather than being confined to a single introduction event. While 4.1% may appear modest, historical patterns from Southeast Asia demonstrate that *kelch13* mutations can reach clinical relevance within a few transmission seasons once established.

Comparison with Published Literature

The near-fixation of *dhfr* triple mutant alleles (N51I + C59R + S108N) observed here (96.5–99.4%) is consistent with findings from neighboring East African countries. Petros et al. [petros2022ethiopia] reported similar *dhfr* triple mutant frequencies exceeding 90% in Ethiopian clinical isolates, while studies across the Horn of Africa document a systematic erosion of SP efficacy. These convergent findings strongly suggest that SP-based IPTp strategies require urgent re-evaluation in the Ethiopian clinical context.

The detection of *kelch13*-P441L at 4.1% is concordant with the sentinel surveillance data compiled by the WHO in 2023 [who2023kelch13], which identified P441L as an emerging validated marker in Eritrea and Ethiopia specifically. Beshir et al. [beshir2022p441l] documented P441L in the Greater Horn of Africa at frequencies between 2–6%, suggesting our findings reflect a regional trend rather than a local anomaly. The geographic distribution across three SRR clusters in our dataset further supports active multi-site dissemination rather than a single introduction event.

Public Health Implications

The co-occurrence of high-frequency SP resistance markers and emerging ACT resistance represents a compound threat to malaria treatment efficacy in Ethiopia. If artemisinin partial resistance continues to increase from its current 4.1% prevalence, Ethiopia may face a scenario analogous to that observed in Southeast Asia, where *kelch13* mutations progressed from marginal frequencies to clinical dominance within five transmission seasons [miotto2015genetic]. Enhanced molecular surveillance programs, ideally integrated with clinical treatment outcome monitoring, should be prioritized to track the trajectory of P441L and any newly emerging *kelch13* mutations in the region.

Limitations

Several limitations of this analysis warrant acknowledgment. First, the amplicon sequencing approach, while powerful for targeted resistance surveillance, provides limited genomic context for population structure inference. The 55 biallelic SNPs identified across ~15 kb are insufficient for robust IBD analysis or genome-wide association studies. Second, geographic metadata linking SRR identifiers to specific clinical sites was not available in the public dataset, limiting the geographic interpretation of P441L distribution. Third, the absence of matched clinical outcome data precludes direct assessment of treatment failure rates associated with the identified mutations.

Conclusions

This independent bioinformatics reanalysis of PRJNA1465284 confirms and extends the molecular epidemiology of antimalarial resistance in Ethiopian *P. falciparum*. The key findings are:

1. The *dhfr* triple mutant conferring SP resistance is near-fixed in clinical isolates from Ethiopian resurgence sites, questioning the efficacy of SP-IPTp in this geographic context.
2. Chloroquine resistance (*pfcr*t-K76T) remains highly prevalent at 81%, consistent with historical trends in East Africa.
3. The artemisinin resistance marker *kelch*13-P441L has been detected at 4.1% across multiple geographic clusters, representing an early but actionable warning signal for ACT efficacy in Ethiopia.

These findings underscore the critical importance of systematic molecular surveillance programs in East Africa and support the urgent need for policy-level response to emerging ACT resistance.

Reproducibility

All analyses were performed on Ubuntu 24.04 LTS (AMD Ryzen 5 PRO, 16 GB RAM, 12 CPUs) using conda/mamba environments. The complete pipeline, including all bash scripts and R code, is publicly available at:

GitHub: github.com/GenoLab-ga

GenoLabGab: genolabgab.vercel.app

Table 7. Software versions used in this analysis.

Tool	Version	Purpose
FastQC	0.12	Per-sample QC
MultiQC	1.x	Aggregate QC report
fastp	0.23	Adapter trimming
BWA-MEM	0.7.18	Reference alignment
samtools	1.x	BAM processing
GATK	4.6.2	Variant calling
SnpEff	5.4	Functional annotation
vcftools	0.1.16	Allele frequencies / diversity
PLINK	1.9	PCA / IBD
R	4.x	Statistical analysis & figures

Acknowledgements

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